



UNITED STATES MILITARY ACADEMY
WEST POINT

Guest Wi-Fi Access

WREN_Visitor

Password: DutyHonorCountry



UNITED STATES MILITARY ACADEMY
WEST POINT®



Chemical Engineering

Advisory Board Meeting

09-10 April 2026

United States Military Academy
Department of Chemical and Biological Science & Engineering



Chemical Engineering

Advisory Board Meeting

10 April 2026

1. Introductions & Opening Remarks

COL Corey James Ph.D., Head CBSE
LTC Victor Jaffett Ph.D., Deputy Head CBSE

United States Military Academy
Department of Chemical & Biological Science & Engineering



- 1500** **Check into Holiday Inn Express & Suites West Point-Fort Montgomery**
- 1106 Route 9w, Fort Montgomery, NY, United States
 - **Contact Dr. Nagelli via call/text upon arrival to hotel at cell: 630-452-3653**
- 1530 – 1630** **Optional Tour of Visitor Center and Walking Tour of Central Area/Post**
(For Board Members Arriving Early; Weather Permitting for Main Post Area)
- [Visitor Center](#) (No Pass needed so can drive direct from Hotel); [Directions](#)
 - Central Area/Trophy Point ([Pass from Visitor Control Center](#) Needed)
 - **Dr. Nagelli/MAJ Rogers with Shuttle at hotel at 1520 for transport to Visitor Center**
- 1630 – 1730** **Obtain visitor pass from Visitor Center (~5-minute processing)**
- Current West Point entry requirements require all non-DOW cardholders to obtain a visitor pass from the Visitor Center. A driver's license, REAL ID, or other valid form of identification is sufficient, and an escort is not required to obtain a pass.
- 1745 – 1800** **Depart Hotel via Shuttle Van (MAJ Rogers & Dr. Nagelli)**
- Optional ride for Board Members or meet at [Restaurant 1915](#)
 - Note there is \$10 parking fee (cash) per vehicle in [Bear Mountain Lot](#)
- 1800 – Until Close** **Dinner at [Restaurant 1915](#), [Bear Mountain Inn](#)**
- Three-course meal with beverages a la carte
 - ChemE Faculty & Board to Attend



0800 – 0830

Session 1: Introduction and Program Assessment

- Introduction & Welcome by COL James
- Program Assessment: PEOs, SOs, and ABET Accreditation (Dr. Nagelli)

0830 – 0930

Session 2: Program Assessment and Objectives

- Program Assessment & Objectives from AY25 (Prof. Biaglow)
- Fundamentals of Engineering Exam Assessment (Dr. Nagelli)
- Advisory Board Feedback from AY24 Program Assessment (Prof. Biaglow)

0930 – 0945

Break

0945 – 1000

Session 3: Introduce Topics for Cadet Panel & Afternoon Small Group Discussions (Dr. Yuk)

1000 – 1010

Advisory Board Moves to Kendrick Aud & Firsties Arrive at 1000

1010 – 1140

Session 4: Board Panel with Firsties & Open Time in Kendrick Aud. (Dr. Yuk)

- What would you change from your experience in the ChemE program?
- Did the Math courses/program support your ChemE courses?
- What was your experience in CH459 ChemE Laboratory Course? What do you think of the course
- Since have unique faculty rotational model, are there programs and resources we can leverage to increase teaching capacity?

1140 – 1150

Group Picture with Cadets, Board Members, and Faculty in Kendrick Aud

1150 – 1200

Cadets Depart to Lunch Formation & Board/Faculty Movement to West Point Club

1200 – 1300

No Host Lunch at Gray Room at West Point Club (Faculty and Board Members)



1310	Return to BH465T Conference Room for Afternoon Session
1310 – 1320	Part I Survey for Board
1320 – 1400	Board Feedback after Panel with Cadets
1400 – 1530	Session 5: <u>Small Groups Workshop with Faculty & Part II Survey for Board (Dr. Nagelli & MAJ Rogers)</u> • Curriculum Updates & Re-design (Senior Design, Bioengineering & Other Specializations) • Industrial & DoD Design Projects and Research Connection • Program Rankings, ChemE Alumni Engagement, Faculty Development and Resourcing
1530 – 1340	Part II Survey for Board
1540 – Until Close	Chemical Engineering Laboratory Tours, Travel back to Hotel if Needed and/or Board Meeting Close Out Social at <u>First Class Club on West Point</u>



1. Advisory Board Members Roles

- **Assessment Data**
- **Objectives Assessment (ABET)**
- **Curricular Challenges/Updates**
- **Complete Surveys**

2. Board Members have discussions with chemical engineering faculty and cadets.

3. Board Members review mission statement, PEOs and future challenges with the anticipated growth of the program/major.



UNITED STATES MILITARY ACADEMY
WEST POINT.



Chemical Engineering

Advisory Board Meeting

10 April 2026

1. Introductions & Opening Remarks

COL Corey James Ph.D., Head CBSE

**United States Military Academy
Department of Chemical & Biological Science & Engineering**



UNITED STATES MILITARY ACADEMY
WEST POINT



Chemical Engineering

Advisory Board Meeting

10 April 2026

Intro to Program Assessment: PEOs, SOs, ABET

Dr. Enoch Nagelli

United States Military Academy

Department of Chemical & Biological Science & Engineering



UNITED STATES MILITARY ACADEMY
WEST POINT®



Chemical Engineering

ABET Accreditation

Accredited **October 01, 2012** to present

Current Record Year: **AY2026**

ABET On-Visit: **August 30 - September 01, 2026**



Engineering
Accreditation
Commission

United States Military Academy
Department of Chemical & Biological Science & Engineering



- Confidence that program has met **standards** essential to prepare graduates to **enter STEM fields**.
- **Keeps us in touch with the engineering profession.**
- Helps USMA (and ChemE) **recruiting** (2020 - 29; 2021 - 20; 2022 - 23; 2023 - 13; 2024 - 21; 2025 - 30; 2026 - 25; 2027 - 28; 2028 - 26; **2029 - 31**).
- **Provides important opportunities for graduates.**
- Allows USMA engineering majors to take the **Fundamentals of Engineering (FE) Exam**.
- **It is required by Army Regulations (10-87).**



Program Educational Objectives

Program educational objectives are broad statements that describe **what graduates are expected to attain within a few years** of graduation.

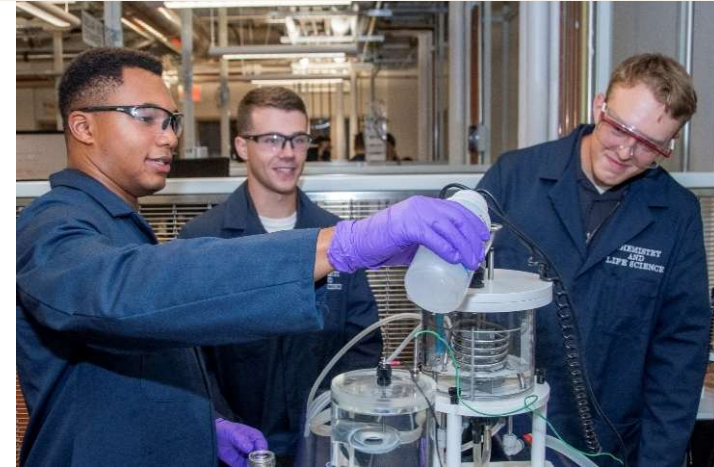
Student Outcomes

Student outcomes describe what students are expected to know and be able to do **by the time of graduation** (skills, knowledge, and behaviors).



Key Terminology

- **Program Educational Objectives (PEOs)**
 - Assembled by asking *program constituents*
 - For us: Army, profession, graduate schools, other
 - **Our external Advisory Board: A key resource.**
 - Desired professional accomplishments of graduates **5-7 years after graduation**
 - Adjust every 3 years or so...
- **Student Outcomes (SOs)**
 - What students should be able to do **at graduation**
 - Must be **measurable**
 - Designed to lead naturally to the PEOs
 - Assess/evaluate some fraction yearly.
- **Assessment → Continuous improvement**
 - Collect meaningful data to evaluate performance indicators (PIs)
 - Assess PIs for outcome attainment → information → COAs for change
 - Implement change
 - Assess its effects and level of success (“closing the loop”)
 - Repeat all the above
 - Periodically check and adjust both Student Outcomes and PEOs





Continuous Improvement (Criterion 4) since 2020 - Assess cadet performance on FE topic or specific graded events:

- **Communication activities in CH367 and CH485.**
- **Addition of and changes to program electives CH300 and CH350.**
- **Changes in course content in EE301.**
- **Changes in course content in CH400.**
- **Introduction of daily economics quizzes in CH402.**
- **Changes in course content in CH365.**
- **Introduction of Mathematica review.**



UNITED STATES MILITARY ACADEMY
WEST POINT



Chemical Engineering

2. Program Assessment and Objectives

Professor Andrew Biaglow

United States Military Academy
Department of Chemical & Biological Science & Engineering



Performance Data

Embedded Indicators
Transcripts
FE Exam

Advisory
Board

Where we are here now!

Collect

Evaluate

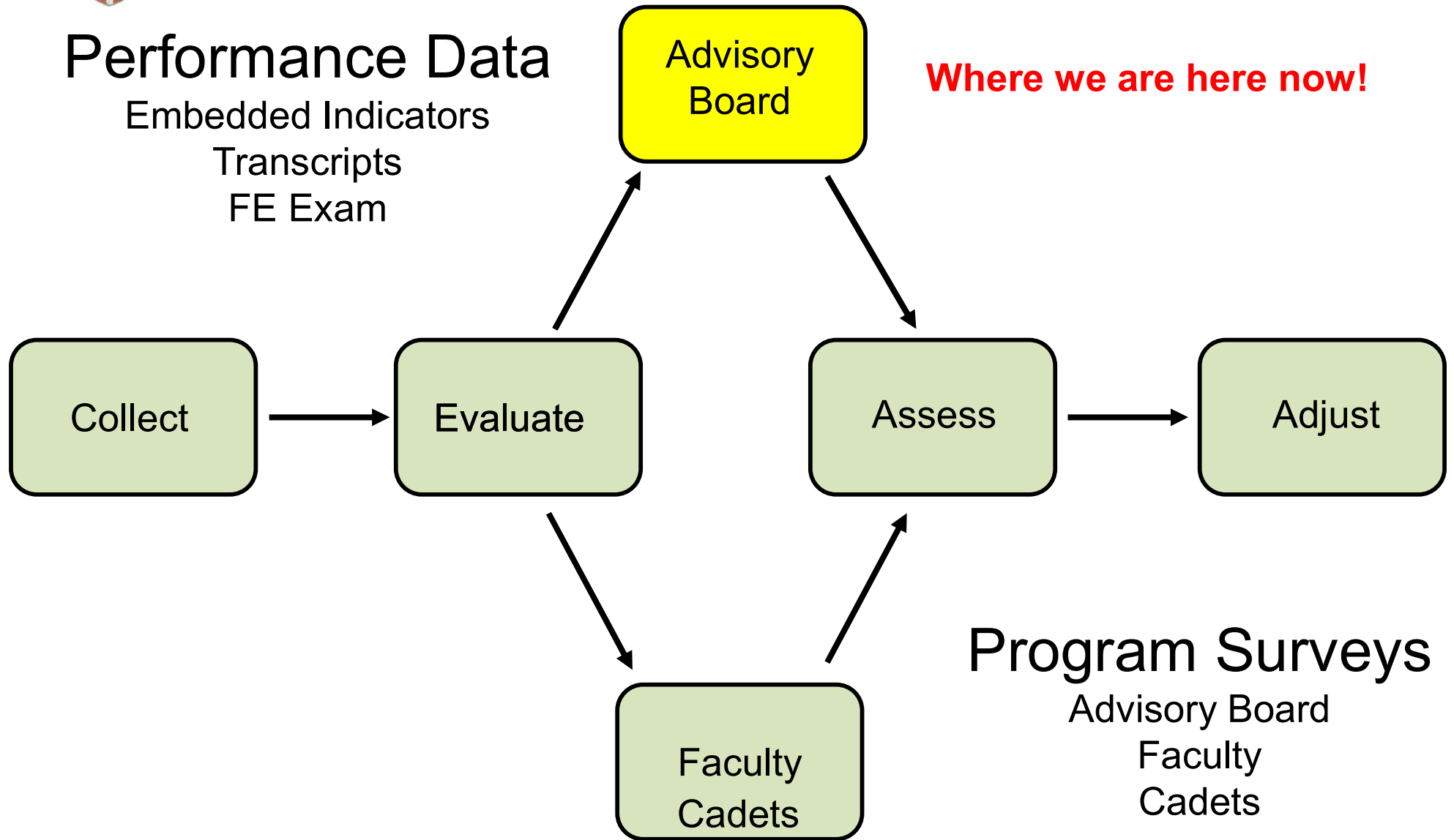
Assess

Adjust

Faculty
Cadets

Program Surveys

Advisory Board
Faculty
Cadets





ABET Student Outcomes (AY19 & beyond)

Identical to ABET 1-7 plus one additional outcome (8)

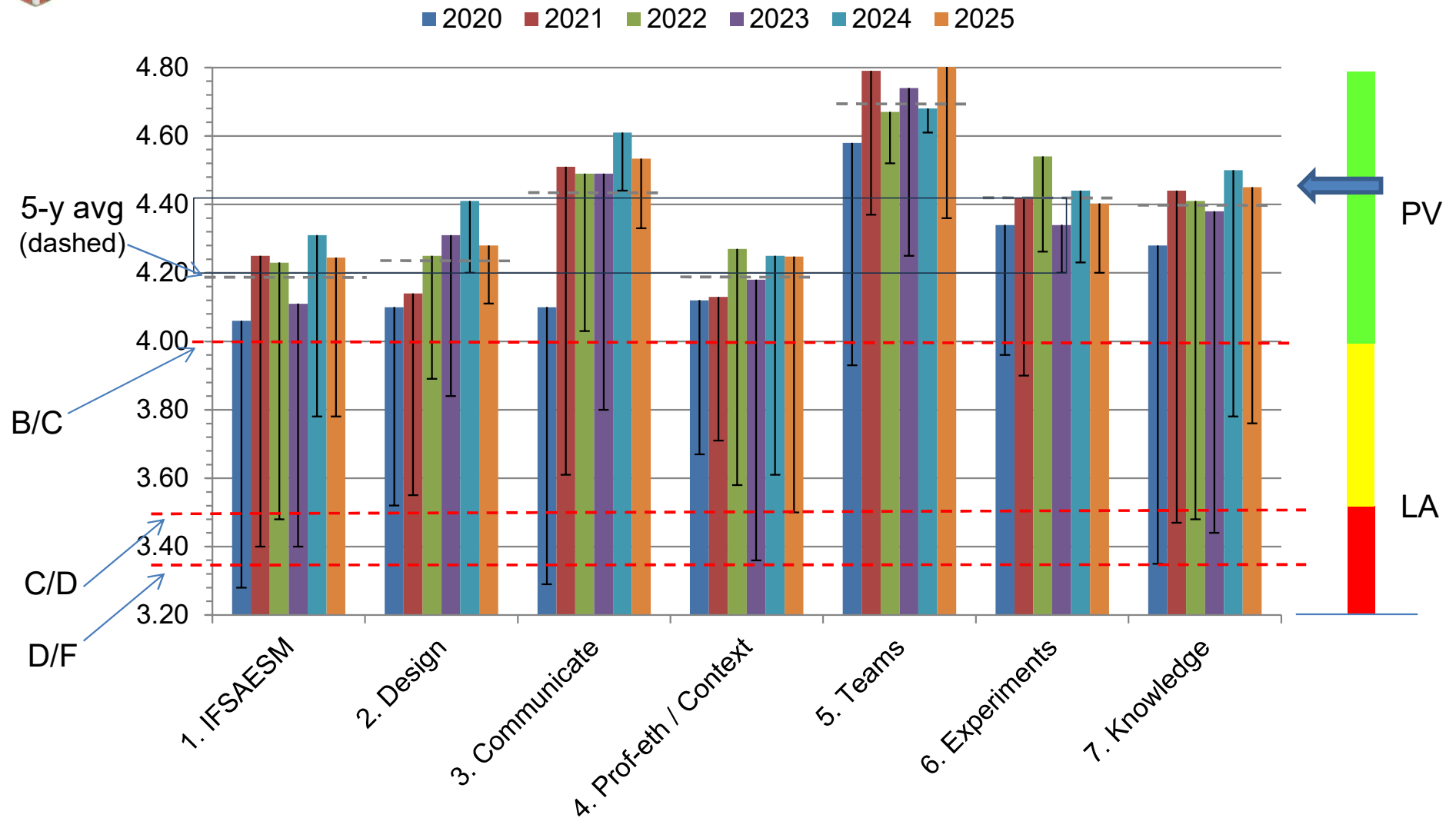
On completion of the chemical engineering program, our graduates will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Communicate effectively with a range of audiences.
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.
8. Understand the chemical engineering curriculum, including chemistry, material and energy balances, safety and environmental factors, thermodynamics of physical and chemical equilibria, heat, mass, and momentum transfer, chemical reaction engineering, continuous and staged separation processes, process dynamics and control, modern experimental and computing techniques, and process design.



Performance on Embedded Indicators

Program Averages AY2020-25

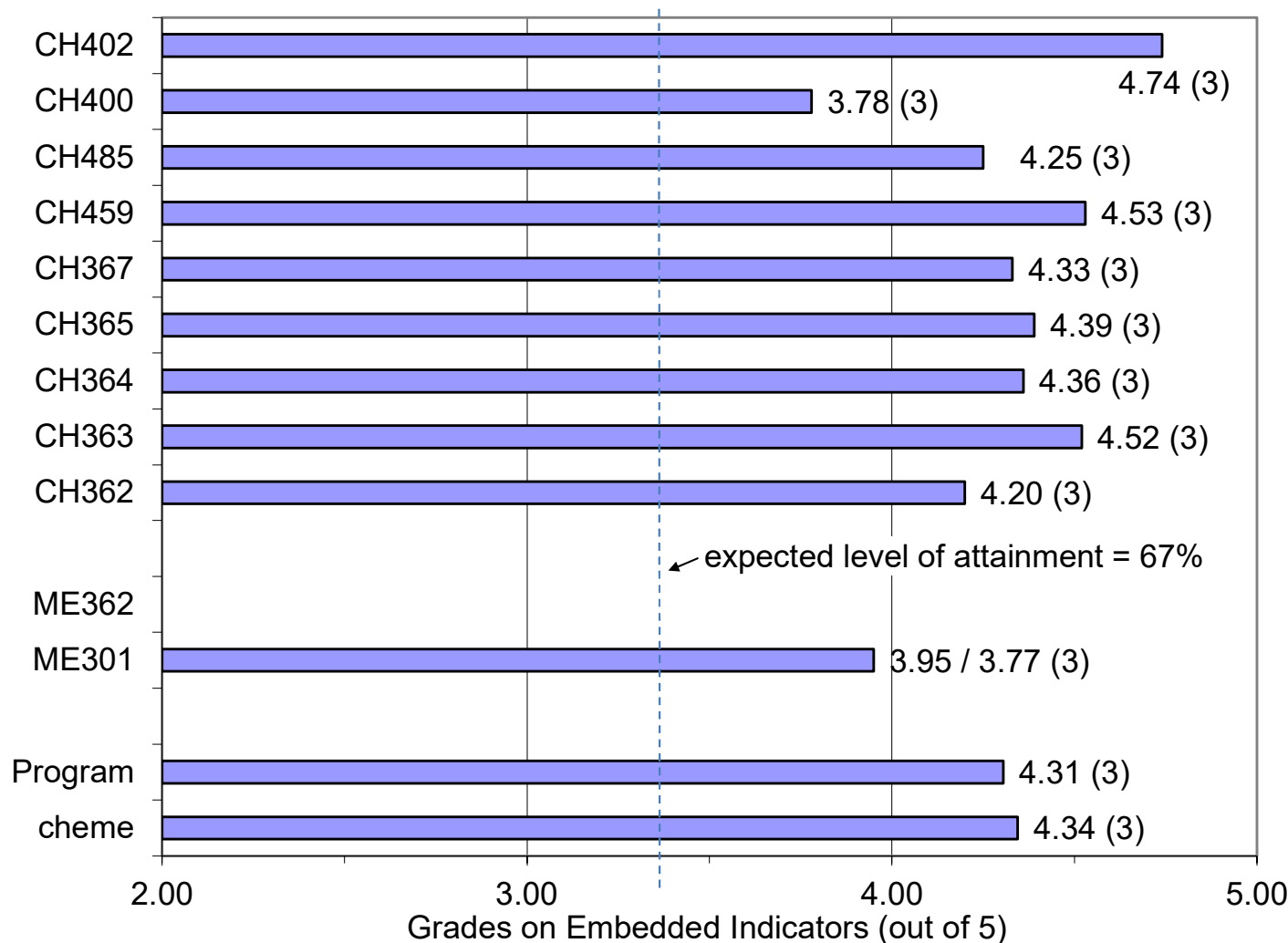


- Expected levels of attainment are the 5-year averages.
- All indicators are in the green and trending upward.

- Cadets are doing very well in all embedded indicators.
- Error bars are minimum scores from course assessments.



Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.



Values in parentheses are coverage ratings from Table 5-4 in the 2020 Self-Study and explained on page 5-7.

Parentheses ratings:.

3: Unique embedded indicator with clear rubric or cut scale.
2: Outcome was graded but grades are convoluted, or part of the outcome is not covered.

1: Correlation to outcome but no assessment.

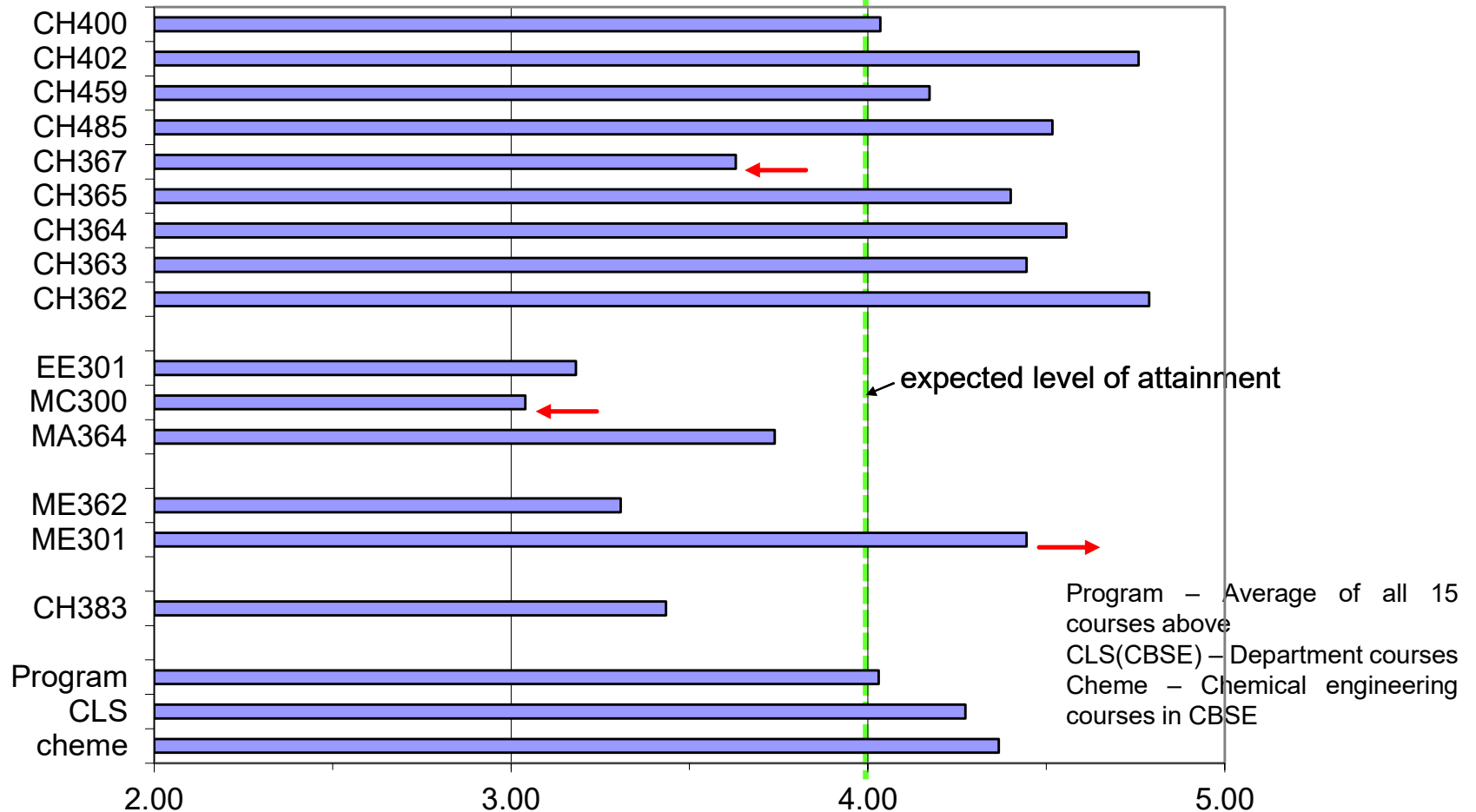
0: No coverage or correlation.

ME301 and ME362 have two scores. The first is for chemical engineering and the second is course-wide.

Data shown here is for Class of 2025
Similar data is collected for all 7 ABET student outcomes
Summary of all data is shown on next slide



This course has improved my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.



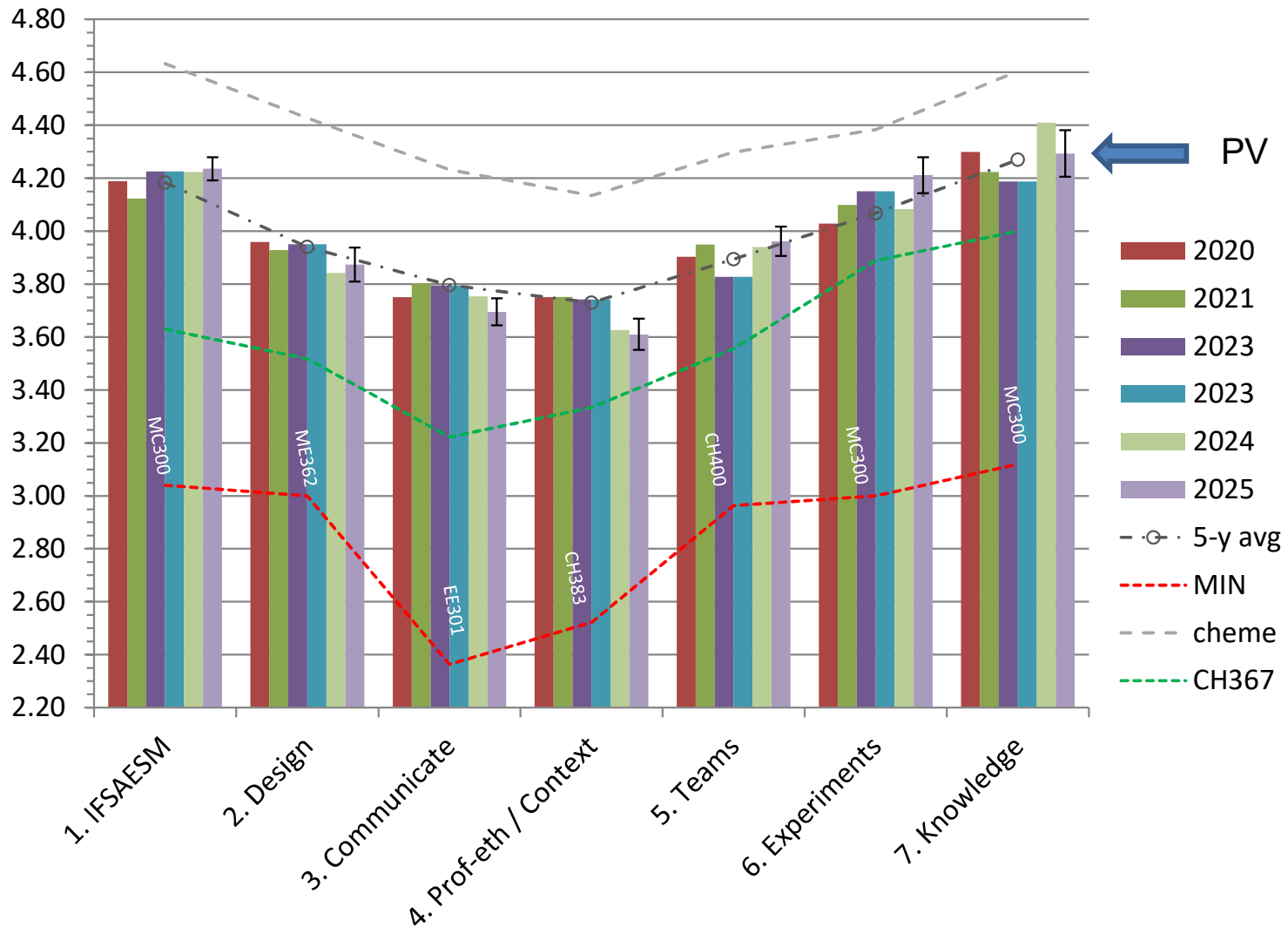
Data shown here is for Class of 2024
Similar data is collected for ABET student outcomes 1-7
Summary of all data is shown on next slide



End-of-Semester Surveys Program Avgs. From AY20-AY25

Error bars are standard deviations.

Expected level of attainment is the 5-y average





Topical Outcomes Evaluation

Student Outcome 8: Understanding of the Chemical Engineering Curriculum

**Average GPA from
Transcripts, AY2020 to
AY2025**

2.4 2.6 2.8 3.0 3.2 3.4 3.6 3.8 4.0

Chemistry

5-year running averages (red solid lines)

8/29

Material & Energy Balances

3/29

Thermodynamics

5/29

Transport Phenomena

5/29

Reaction Engineering

5/29

Separations

2/29

Dynamics & Control

4/29

Experimental & Computing Tools

0/29

Process Design

4/29

Expected level of attainment for the program is B- (2.67, red dotted line).

AY20

AY21

AY22

AY23

AY24

AY25

Letter Grade Grade Point Conversion

A+ 4.33

A 4.00

A- 3.67

B+ 3.33

B 3.00

B- 2.67

C+ 2.33

C 2.00

C- 1.67

D 1.00

F 0.00



UNITED STATES MILITARY ACADEMY
WEST POINT.

Break

0935-0950



Chemical Engineering

2. Program Assessment and Objectives (continued)

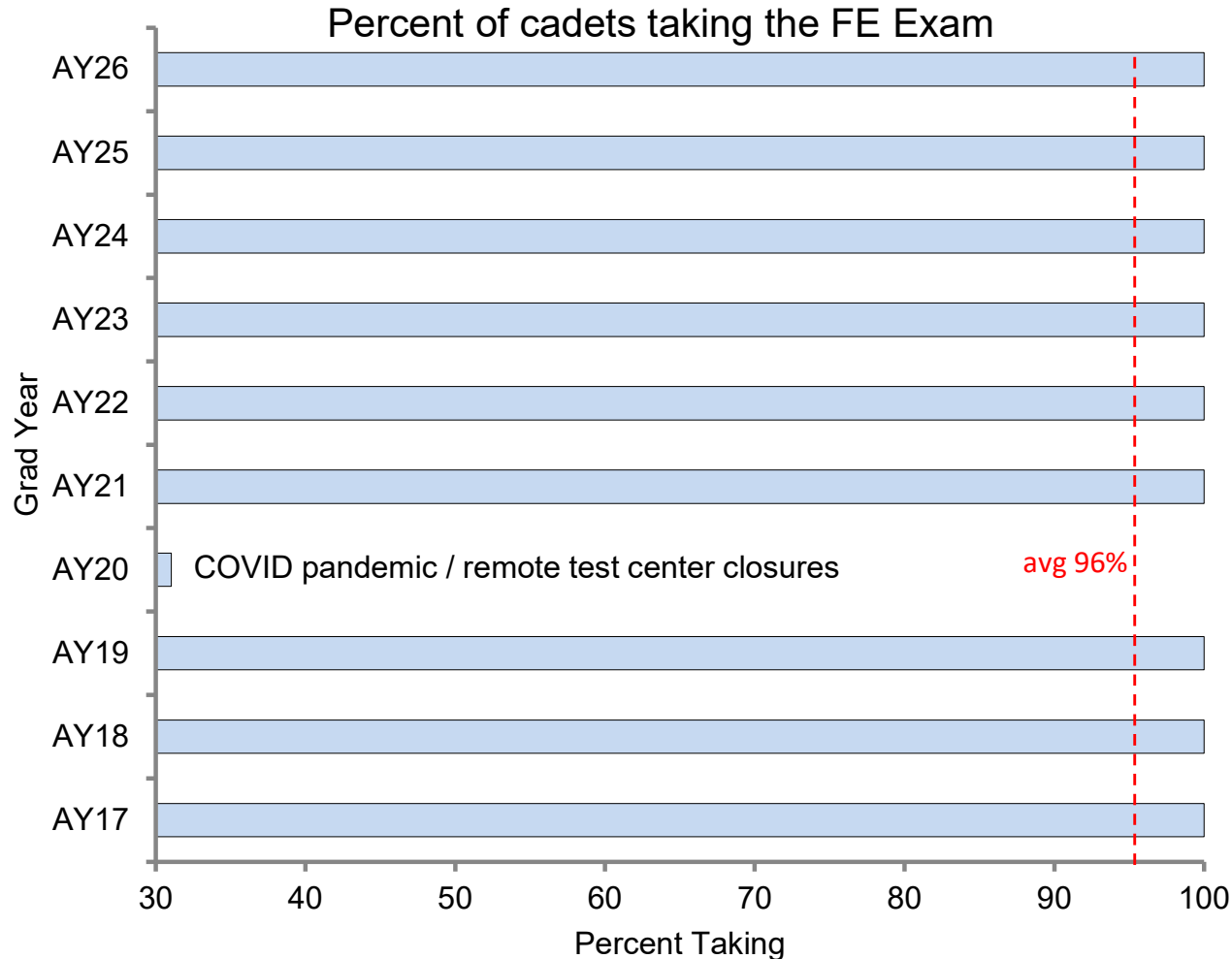
Fundamentals Engineering Exam Results

Dr. Enoch Nagelli

**United States Military Academy
Department of Chemical & Biological Science & Engineering**



Student Outcome 7: Acquire and apply new knowledge as needed, using appropriate learning strategies

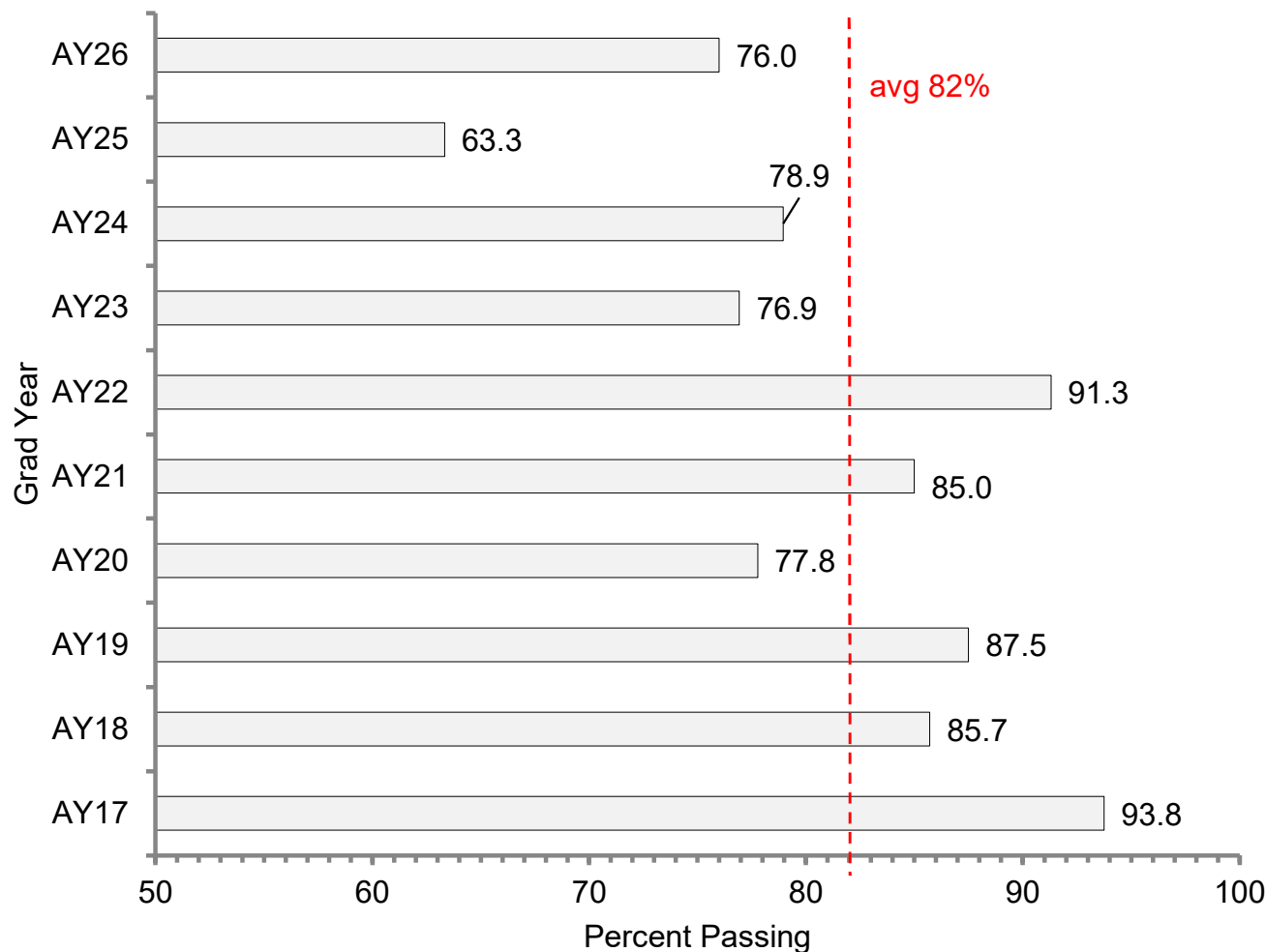


The average take rate was 86% prior to introduction of CH400 in AY2010.



Student Outcome 7: Acquire and apply new knowledge as needed, using appropriate learning strategies

Percent of cadets passing the FE Exam



National, % Pass:

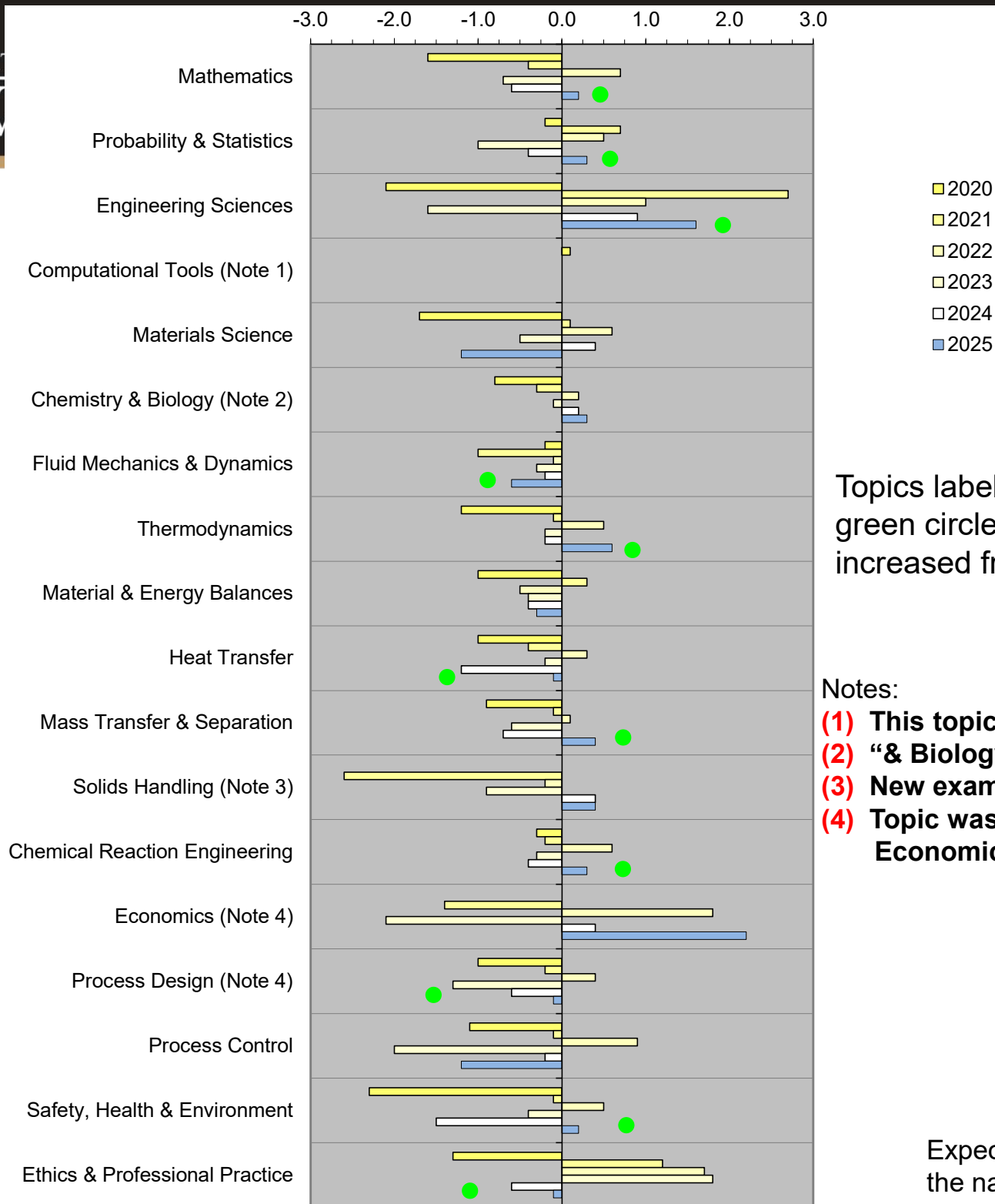
2026	71.0%*
2025	70.9%
2024	73.2%
2023	70.2%
2022	70.7%
2021	74.0%
2020	74.6%
2019	77.0%
2018	75.0%
2017	74.0%
2016	79.0%

*estimated



Topical Outcomes Evaluation

Deviations from National Averages AY20 to AY25



Expected level of attainment is the national average (0.0 line)



Chemical Engineering

2. Program Assessment and Objectives (continued)

Advisory Board Feedback from 2025

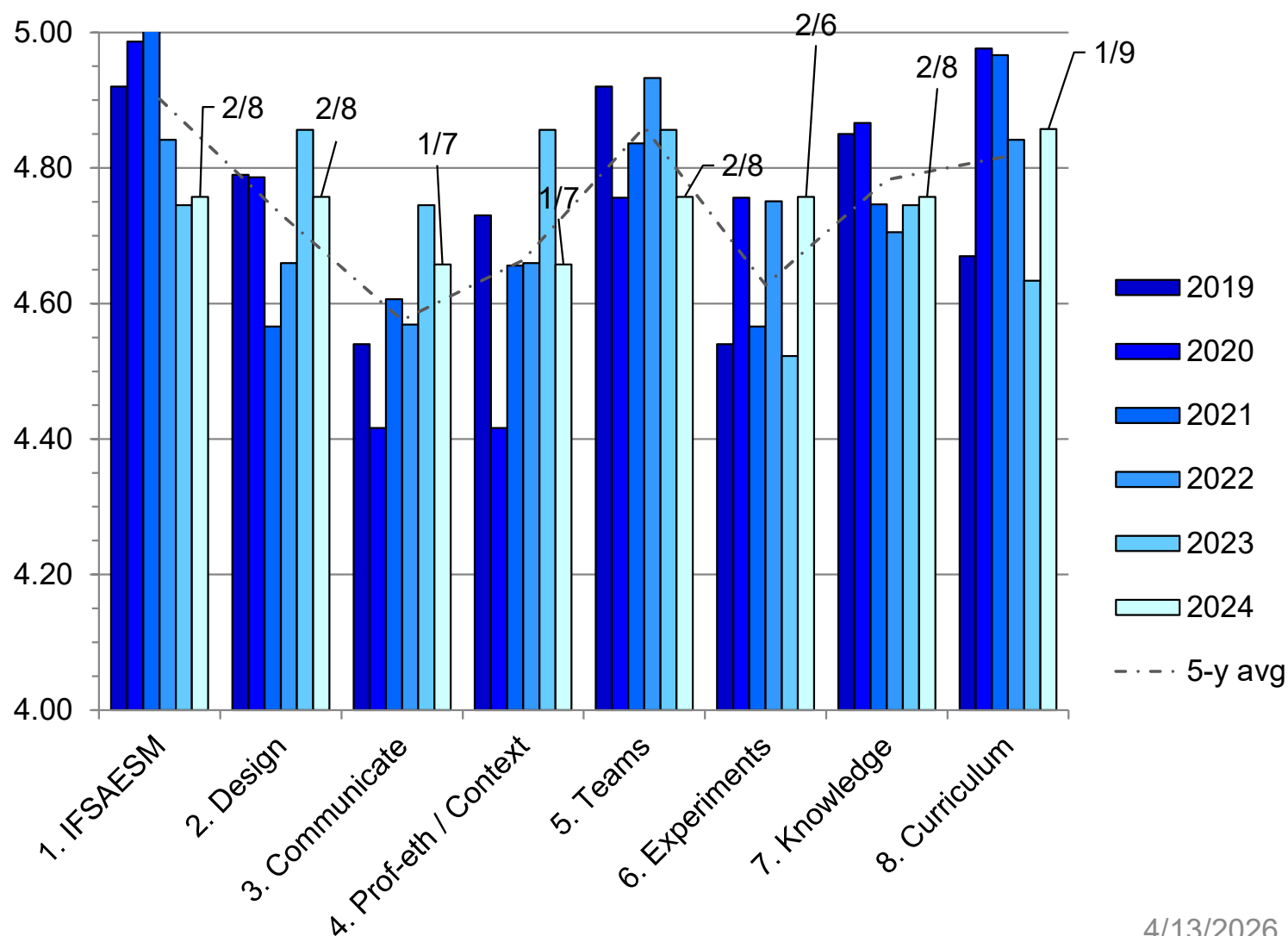
Prof. Andrew Biaglow

**United States Military Academy
Department of Chemical & Biological Science & Engineering**



Program Averages from AY19-24

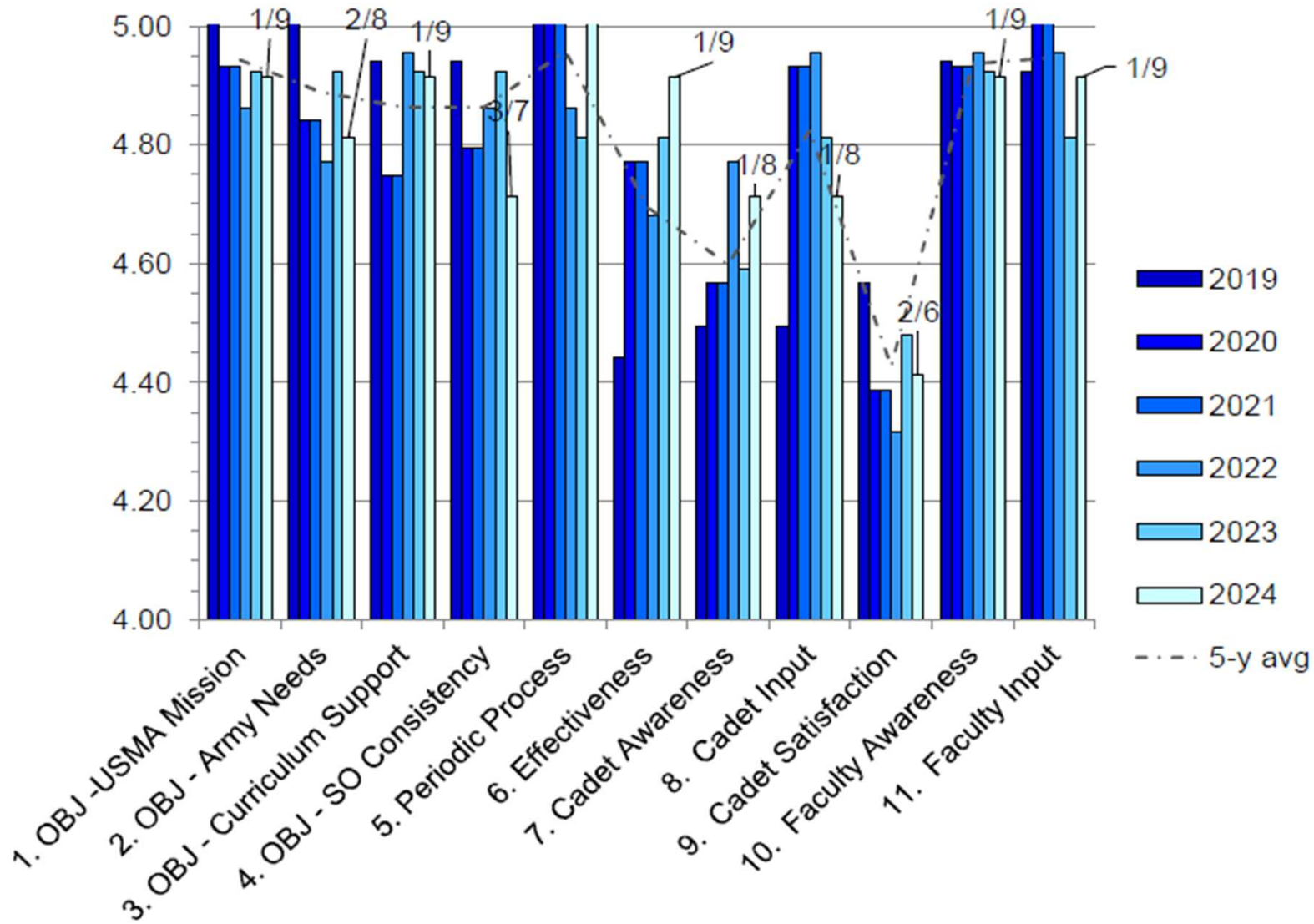
Data labels are response frequencies for 4 or 5 (# of 4s / # of 5s) on the 1-5 Survey Likert Scale
Standard deviations range from .00 to .52





Program Averages from AY19-24

Data labels are response frequencies on the 1-5 Survey Likert Scale (# of 3 / # of 4 / # of 5).





Summary from Surveys Completed Remote as of 03 June 25

Note: In-person Advisory Board Meeting Cancelled due to DoW guidance for the reduction and consolidation of Government travel

- The program is well-rounded and has the right courses. Suggestions for additional courses are numerical methods, organic 2, or some sort of bio course.
- Advisory board members are concerned about the lack of an in-person meeting.
- In-person meetings are essential for meaningful program evaluation due to the unique insight gained through personal interaction with cadets and faculty.
- Decline in cadet performance, specifically highlighting a two-year drop in FE exam pass rates and Organic Chemistry grades, and a dip in ethics and safety knowledge.
- Correlation between GPA and FE results, investigating the mismatch between FE prep course surveys and embedded student indicators.



UNITED STATES MILITARY ACADEMY
WEST POINT.



Chemical Engineering

3. Introduce Cadet Panel Topics & Afternoon Small Group Discussion Topics

Dr. Simuck Yuk

**United States Military Academy
Department of Chemical & Biological Science & Engineering**



- Program **Sustains/Improves**
- Any **general issues** with the program you would like to discuss?
- Any thoughts on **fundamental courses** that prepare you for upper level courses?
- What are your thoughts on **structure of labs**?
- What do you think the **honors requirement** for graduation should be?
- Are you happy with your **elective choices**?
- What was your **FE Exam** experience like?
- If you were going into the job market as a new engineer, **do you feel prepared**?



Future Curriculum Changes

- Does current curriculum prepare students for the current and future landscape of ChemE to include the emerging fields?
 - Do we have the correct compliment of major courses?
 - New courses or specializations relevant to industry trends

Strengthening Program Connections with Industry/Army Needs

- How can we strengthen connections with industry to provide more real-world projects and mentorship opportunities?
 - Capstones tailored towards Army/Industry-focused efforts
 - Research Projects addressing specific Army needs

Undergraduate Rankings and Program Alumni Engagement

- What are key aspects of the program that impact overall rankings?
- How can we incorporate feedback from alumni and industry partners to continuously improve our curriculum?
- Connect cadets with alumni in the Army



UNITED STATES MILITARY ACADEMY
WEST POINT®



Chemical Engineering

4. Advisory Board Survey (Part 1)

Dr. Enoch Nagelli

United States Military Academy
Department of Chemical & Biological Science & Engineering



UNITED STATES MILITARY ACADEMY
WEST POINT.

Advisory Board Completes Survey Part 1



UNITED STATES MILITARY ACADEMY
WEST POINT



Chemical Engineering

5. Small Groups Workshop with Faculty

Dr. Nagelli & MAJ Rogers

United States Military Academy
Department of Chemical & Biological Science & Engineering



Curriculum Workshop (1400-1500)	Room Location Assigned Board Members & Faculty
Group 1: Curriculum gaps, overlaps, & priorities - Where is the curriculum overloaded or redundant? - What are the biggest gaps relative to industry/Army needs? - Within the USMA constraints of 4 years, what changes to the required major courses would you make?	Conference Room LTC(R) Matthew Armstrong, PhD Dr. Kelly Schultz Dr. Enoch Nagelli MAJ Josh Frey
Group 2: Labs, Design and Experiential Learning - Does the Unit Operations Lab (Senior Lab) Design Course meet the intent? - How should our current capstone design/projects in each course evolve?	BH331 COL(R) Paul Dietrich Dr. Gautham Krishnamoorthy Mr. Michael DeForest Dr. Andy Biaglow MAJ Samuel Lowell
Group 3: Digital & Data Skills in ChemE - What tools should every graduate know (e.g., Python, process simulators)? - Should intro to coding methods be a standalone required course?	BH341B Dr. Susan Daniel Mrs. Kisondra Tanev Mr. Michael Theising Dr. Simuck Yuk MAJ Louis Tobergte
Group 4: Enhance Faculty Development & Resourcing Considering USMA's unique faculty rotational model, what strategies or efforts can the program leverage to implement to enhance faculty development?	BH341A COL Aaron Hill, PhD, PE Dr. Matthew Liberatore Mr. Kevin Shipe MAJ Nijel Rogers

ALL GROUPS: Undergraduate **rankings in ChemE** & Alumni Engagement

Out-brief from Each Group (1500-1530): Top Recommendations from each group with one high impact change. Highlight timeframe: Immediate (1-2 years) or Strategic (3-5 years)



UNITED STATES MILITARY ACADEMY
WEST POINT®



Chemical Engineering

6. Advisory Board Survey (Part 2)

Dr. Enoch Nagelli

United States Military Academy
Department of Chemical & Biological Science & Engineering



Advisory Board Completes Survey Part 2



- For the opportunity to show you America's Military Academy
- For your service and insights to help our program improve
- For the time you have dedicated to this visit
- For your dedication to the profession





- Next Advisory Board at USMA Early/Mid April 2027
- Travel Paperwork/Receipts
 - DTS Voucher Mrs. Kristen Costain
- Tour of Unit Operations Lab and BH labs...UTC
- Shuttle back to Hotel – Pick up in front of BH (~1615)
- Close Out Social at Firstie Club at 1630.

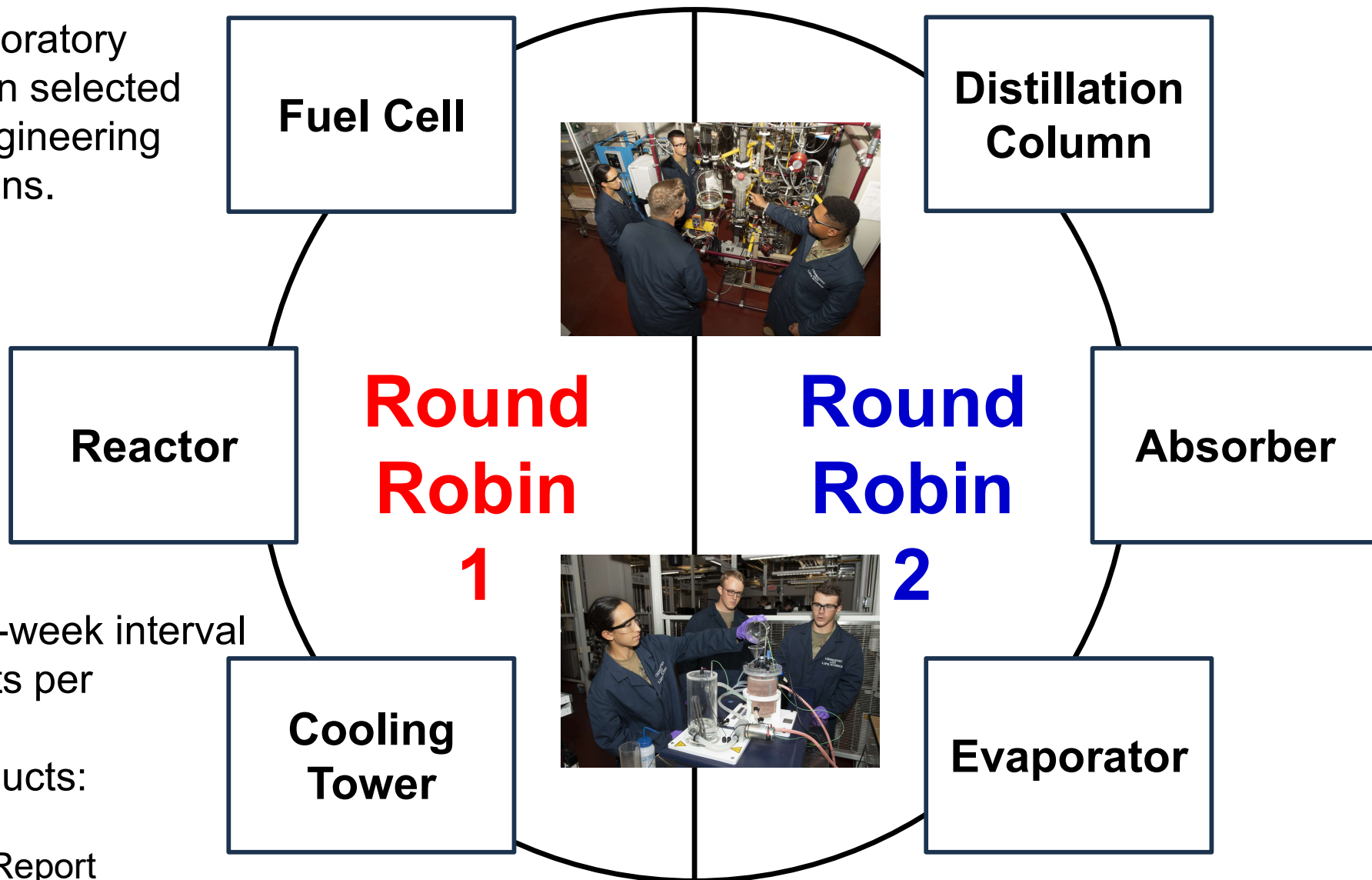


UNITED STATES MILITARY ACADEMY
WEST POINT.

Extra Slides



Provides laboratory experience in selected chemical engineering unit operations.



Rotates at 2-week interval
(4 to 5 cadets per experiment)

Written Products:

- Poster
- Technical Report
- Executive Summary



Chemical Engineering

United States Military Academy

Example Schedule for Chemical Engineering, Classes of 2024 and Beyond



Fall Term	Course	Credit Hours	Spring Term	Course	Credit Hours
4th CLASS					
MA103	Math. Modeling & Intro. Calculus	4.5	MA104	Calculus I	4.5
CH101	General Chemistry I	4.0	CH102	General Chemistry II	4.0
EN101	Composition	3.0	EN102	Literature	3.0
HI101	The Army of the Republic	3.0	HI108	Regional Studies in World History	3.0
CY105	Introduction to Computing & IT	3.0	PL100	General Psychology	3.0
PE11x	Combatives / Boxing / Movement	0.5	MS100	Introduction to Warfighting	1.5
			PE150	Fundamentals/Personal Fitness	1.5
3rd CLASS	Total	18.0		Total	20.5
MA205	Calculus II	4.0	CH362	Mass and Energy Balances	3.5
PH205	Physics I	4.0	MA364	Engineering Mathematics	3.0
Lx203	Foreign Language	4.0	PH206	Physics II	4.0
SS201	Economics	3.0	Lx204	Foreign Language	4.0
PY201	Philosophy	3.0	SS202	American Politics	3.0
MS200	Fundamentals: Army Operations	1.5	EV203	Physical Geography	3.0
			PE 2xx	Lifetime Physical Activity	0.5
2nd CLASS	Total	19.5		Total	21.0
CH363	Separation Processes	3.5	CH364	Chemical Reaction Engineering	3.5
EE301	Fundamentals of Electrical Engineering	3.5	CH367	Introduction to Automatic Process Control	3.0
CH383	Organic Chemistry I	3.5	ME362	Fluid Mechanics	3.5
ME301	Thermodynamics	3.5	MC300	Fundamentals of Eng. Mech. & Design	3.0
PL300	Military Leadership	3.0	SS307	International Relations	3.0
MA206	Probability and Statistics	3.0	MS300	Platoon Operations	1.5
PE32x	Survival Swimming	0.5	PE360	Combat Applications	1.5
1st CLASS	Total	20.5		Total	19.0
CH459	Chemical Engineering Laboratory	3.5	CH402	Chemical Engineering Process Design	3.5
CH365	Chemical Engineering Thermodynamics	3.0	CH400	Chemical Engineering Professional Practice	1.5
CH485	Heat & Mass Transfer	3.5	Elective	Engineering Elective 3	3.0
Elective	Engineering Elective 1	3.0	HI302	History of the Military Art	3.0
Elective	Engineering Elective 2	3.0	LW403	Constitutional & Military Law	3.0
PE450	Army Fitness Development	1.5	MX400	Officership	3.0
	Total	17.5		Total	17.0



**Pre-approved elective sequences,
but ultimately cadet choice (can choose any 3)**

Materials Engineering

MC364 Mechanics of Materials

MC380 Engineering Materials

Open Elective

Nuclear Engineering

NE300 Nuclear Reactor Analysis

NE350 Nuclear Reactor Design

NE450 Nuclear Systems Design

Decision Analysis

SE301 Foundations of Engineering Design
& Systems Management

SE 481 Systems Simulation

EM484 Dynamic Systems Analysis

Advanced Control Systems

EE360 Digital Computer Logic

EM484 Dynamic Systems Analysis

XE475 Mechatronics

Energy Conversion Systems

EE377 Electrical Power Generation

ME472 Energy Conversion Systems

ME480 Heat Transfer

Power Systems

ME306 Dynamics

ME491 Mechanical Power Plants

EE377 Electrical Power Generation

XE442 Alternative Energy Engineering

Industrial Engineering

SE301 Foundations of Engineering Design
& Systems Management

EM411 Project Management

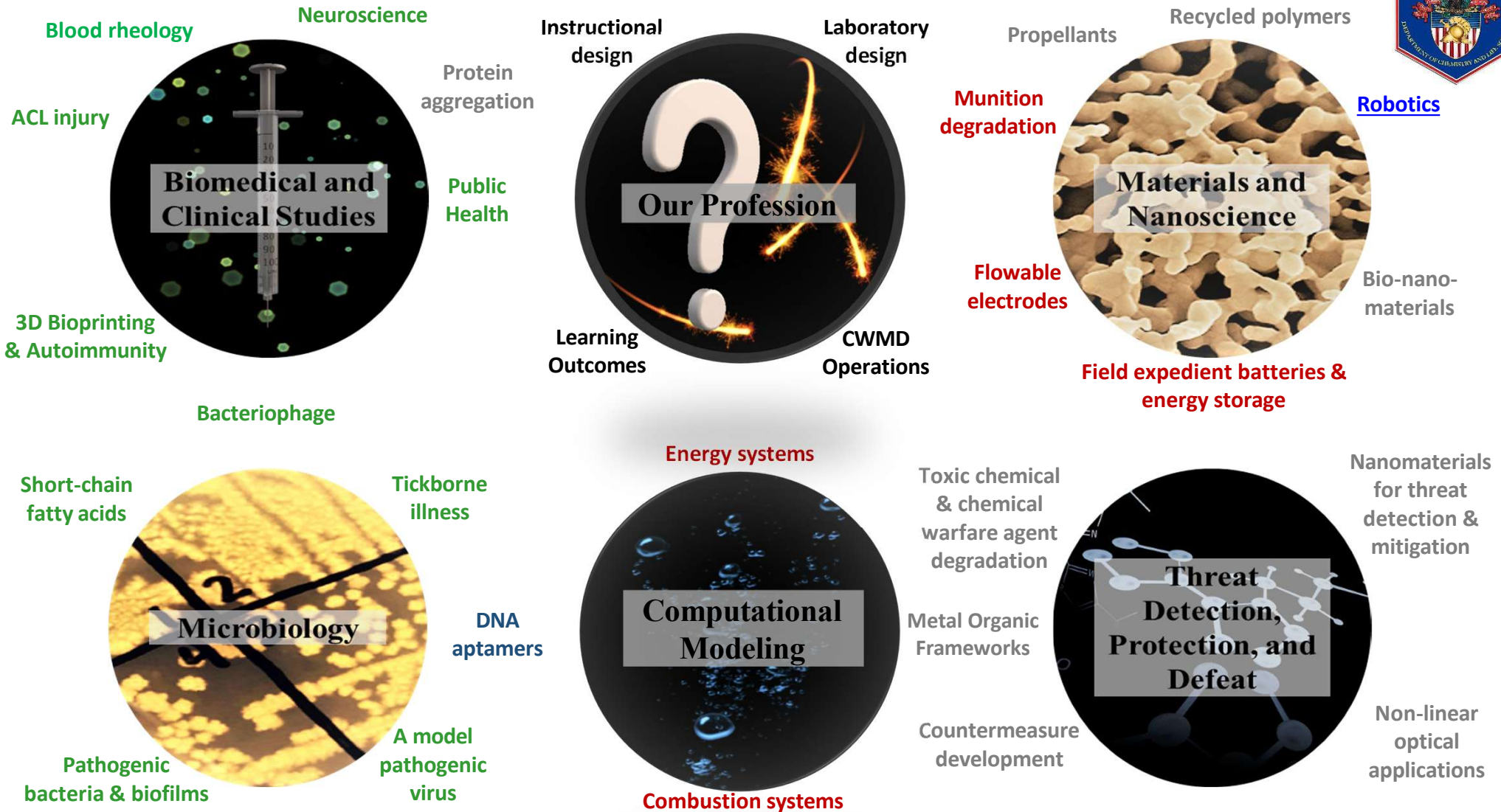
EM420 Production Operations Management

Other Advanced Engineering Electives

Satisfy prerequisites

Engineering Science or design = 3.0 credits

Program director approval



Program Lead: **Bioengineering** **Chemical Engineering** Chemistry **Biology**



Research Model

Developmental Opportunities Afforded by Faculty Research Collaboration with an Army/DoD Laboratory

Cadet Development

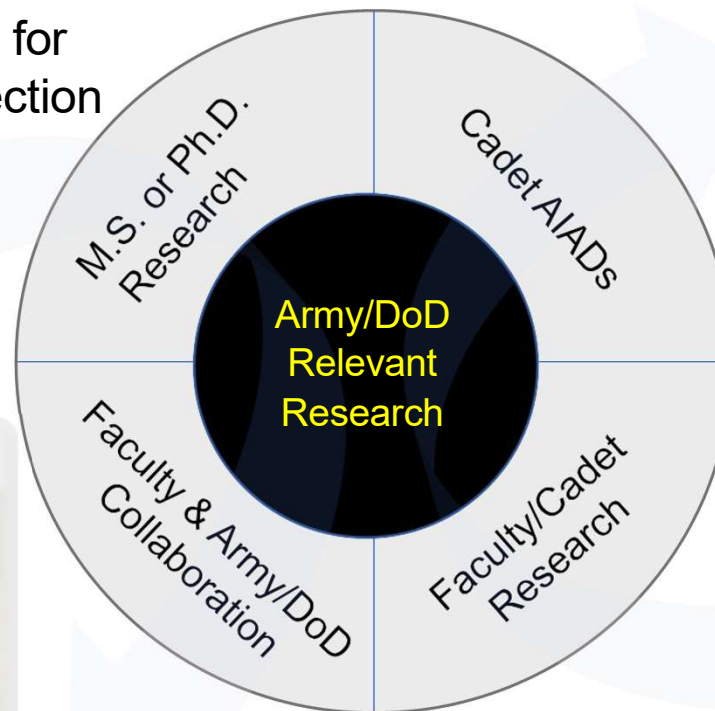
Mentoring for
School Selection

Faculty Selection
into a Discipline

Recruiting

Intellectual
Capital

Faculty Development



Return to
Operational
Force

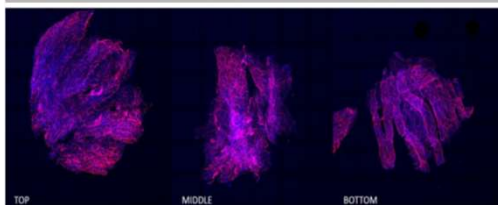
Coursework and
Interest



Every interaction is a developmental event



Biomedical & Clinical Studies



3D Bioprinting
&
Autoimmunity

Tickborne
illness

Non-linear
optical
applications

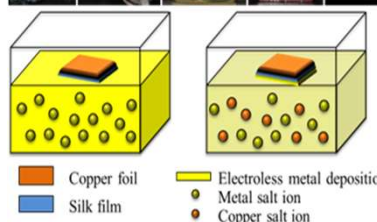
Toxic chemical
& chemical
warfare agent
degradation

DNA
aptamers

Bacteriophage
therapeutics

ACL injury

Bioengineering

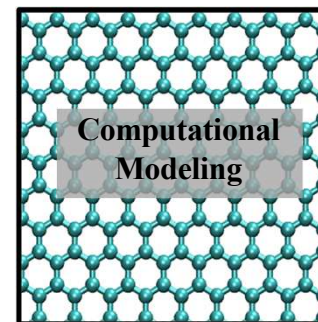


Protein
aggregation

Pathogenic
bacteria and viruses

Neuroscience

Public
Health



Computational
modelling

AI/ML soft
sensors

Materials

Nanomaterials
for threat
detection &
mitigation

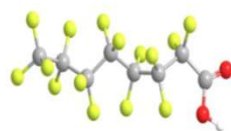
Countermeasure
development

Metal Organic Frameworks

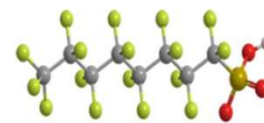
Robotics

Flowable electrodes

Bio-nano-
materials



Structure of PFOA⁵



Structure of PFOS⁶

Field expedient batteries & energy
storage

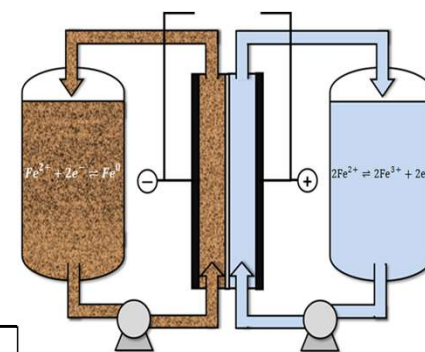
STEM
education

Per- and Poly-
Fluoroalkyl
Substances (PFAS)

Energy Storage

Infrastructure
process
optimization

Combustion
systems



Munition degradation

Renewable energy

Propellants

Sustainability

Program Lead: **Chemical Engineering** Chemistry **Biology**

